

Dynamics and relativity

OLIVER WEST

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§1 The structure of the Newtonian universe

- Three dimensional space endowed with a *Cartesian reference frame* – an origin and some axes.
- Time can be labelled in an arbitrary reference frame, by a real number t .
- A *point particle* is an idealised object that is completely determined by its position at a given point in time, $\mathbf{x}(t)$. Its velocity

$$\mathbf{v} = \frac{d\mathbf{x}}{dt} = \left(\frac{dx_1}{dt}, \frac{dx_2}{dt}, \frac{dx_3}{dt} \right) = \dot{\mathbf{x}}$$

is tangent to the trajectory. The acceleration

$$\mathbf{a} = \frac{d^2\mathbf{x}}{dt^2} = \left(\frac{d^2x_1}{dt^2}, \frac{d^2x_2}{dt^2}, \frac{d^2x_3}{dt^2} \right) = \ddot{\mathbf{x}} = \dot{\mathbf{v}}$$

The above is *not* sufficient to write down Newton's equations. Consider, for example, a free particle not experiencing any forces. The position of this particle is $\mathbf{x}(t)$, but which frame should we use?

§1.1 Law of inertia

Proposition 1.1 (Law of inertia)

There exist *inertial frames* in which a free particle has a constant velocity.

This is an improved version of Newton’s first law. Indeed, it is a true statement, but not an obvious one.

§1.2 Galilean relativity principle

Definition 1.2. A *Galilean transformation* has the form

$$\mathbf{x}' = R\mathbf{x} + \mathbf{k} + \mathbf{w}t$$

where $R \in O(3)$, $\mathbf{k}$ is a translation vector, and $\mathbf{w}$ is a *boost*. In particular, $\ddot{\mathbf{x}}' = \ddot{\mathbf{x}}$.

Proposition 1.3 (Galilean relativity principle)

A frame related to an inertial frame by a *Galilean transformation* is also an inertial frame, and all laws of physics are the same in both frames.

Example of a boost: dropping a ball on a boat moving at constant velocity.

Galilean invariance also restricts the forces that are possible (see example sheet). For example, there are no ‘special points’ in the universe (or special directions, times, velocities, etc) – they are all relative. This is *not* the case for acceleration – an object accelerating in one inertial frame is accelerating with the same magnitude in all inertial frames.

Galilean transformations form the *Galilean group*, often supplemented by shifts in time, under which the laws of physics are also invariant. In particular, this implies that all inertial frames have the same time, called absolute time.

§2 Forces

Once there is more than one particle in the universe, there will be interactions between them. In Newtonian physics, these are described by forces.

§2.1 Newton’s second law

In an inertial frame,

$$\dot{\mathbf{p}} = \mathbf{F}$$

where the *momentum* $\mathbf{p}$ is $m\mathbf{v}$, where m is the *inertial mass*, mass for short. The mass is an additional property for point particles that we take to be constant in time unless otherwise stated.

The force $\mathbf{F}$ depends on the ongoing interactions, but it can only depend (axiomatically) on $\mathbf{x}$ and $\mathbf{v}$. This implies that Newton’s second law is a second-order differential equation, and that, given $\mathbf{x}$ and $\mathbf{v}$ at $t = 0$ for all particles, Newton’s equations will uniquely determine $\mathbf{x}(t)$ for all future times.

We should note that Newtonian mechanics has been superseded by quantum mechanics (for small things) and general relativity (for fast things).

§2.2 Conservative forces

Definition 2.1. A force $\mathbf{F}$ is called *conservative* if it can be written

$$\mathbf{F} = -\nabla V$$

for some *potential* scalar field V .

For example, *gravitational potential energy* of a particle of mass m at $\mathbf{x}$ due to a particle of mass M at $\mathbf{x}_0$ is

$$V = -\frac{GMm}{|\mathbf{x} - \mathbf{x}_0|}$$

where G is a constant.

Now we can get the force by taking the gradient. First,

$$\partial_i (|\mathbf{x} - \mathbf{x}_0|^2) = 2|\mathbf{x} - \mathbf{x}_0| \partial_i |\mathbf{x} - \mathbf{x}_0|$$

But also,

$$\begin{aligned} \partial_i (|\mathbf{x} - \mathbf{x}_0|^2) &= \partial_i ((\mathbf{x} - \mathbf{x}_0)_j (\mathbf{x} - \mathbf{x}_0)_j) \\ &= 2(\mathbf{x} - \mathbf{x}_0)_j \partial_i (x_j - (x_0)_j) \\ &= 2(\mathbf{x} - \mathbf{x}_0)_j \delta_{ij} \\ &= 2(\mathbf{x} - \mathbf{x}_0)_i \end{aligned}$$

and so we have

$$\begin{aligned} \partial_i |\mathbf{x} - \mathbf{x}_0| &= \frac{(\mathbf{x} - \mathbf{x}_0)_i}{|\mathbf{x} - \mathbf{x}_0|} \\ \Rightarrow \nabla |\mathbf{x} - \mathbf{x}_0| &= \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|} \end{aligned}$$

Hence

$$\mathbf{F} = -\nabla V = -GMm \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|^3}$$

Let $\mathbf{r} = \mathbf{x} - \mathbf{x}_0$, then

$$\mathbf{F} = -\frac{GMm}{r^2} \hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is a unit vector. Thus we have the inverse square law of gravity.

Remark. Sometimes we write $V = m\Phi$ where $\Phi = -\frac{GM}{|\mathbf{x} - \mathbf{x}_0|}$ is called the *gravitational potential*.

Near the surface of the Earth, take $\mathbf{x}_0 = \mathbf{0}$, and $|\mathbf{x}| = R + z$, where R is the radius of the Earth and $z \ll R$. Then

$$\begin{aligned} \Phi(R + z) &= -\frac{GM}{R + z} \\ &= -\frac{GM}{R} \left[1 - \frac{z}{R} + \frac{z^2}{R^2} + \cdots \right] \end{aligned}$$

$$\approx \text{constant} + \underbrace{\frac{GM}{R^2}}_g z$$

where $g \approx 9.8\text{ms}^{-2}$. Since the constant will drop out when taking the gradient, we ignore it, so we have

$$\begin{aligned}\Phi &\approx gz \\ \implies \mathbf{F} &= -mg\hat{\mathbf{z}}\end{aligned}$$

This force leads to the simplest non-trivial example of motion due to a force. By Newton's second law, we have

$$m\ddot{\mathbf{x}} = m\mathbf{g}$$

where $\mathbf{g} = (0, 0, -g)$. We need only consider the z component, which gives

$$\begin{aligned}m\ddot{z} &= -mg \\ \implies \dot{z} &= v_0 - gt \\ \implies z &= z_0 + v_0t - \frac{1}{2}gt^2\end{aligned}$$

for an initial height z_0 and an initial velocity v_0 .

§2.3 Energy

Proposition 2.2

Conservative forces have a *conserved energy*

$$\begin{aligned}E &= \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + V(\mathbf{x}) \\ &= \frac{1}{2}m\dot{x}_i\dot{x}_i + V(\mathbf{x})\end{aligned}$$

Proof. We have

$$\begin{aligned}\frac{dE}{dt} &= m\dot{x}_i\ddot{x}_i + \frac{\partial V}{\partial x_i}\dot{x}_i \\ &= m\dot{x}_i \left(m\ddot{x}_i + \frac{\partial V}{\partial x_i} \right) \\ &= 0\end{aligned}\quad (m\ddot{\mathbf{x}} = -\nabla V)$$

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Example 2.3

Suppose we throw an object into space and want it to never fall back down. The minimal velocity required is called the escape velocity.

When the object is thrown, its energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

To not fall back, the object must reach $|\mathbf{x}| \rightarrow \infty$ without the velocity going to zero. We have

$$E_\infty = \frac{1}{2}mv_\infty^2 - 0$$

By conservation of energy, $E = E_\infty$, and so to make $v_\infty > 0$ we must have

$$\begin{aligned} \frac{1}{2}mv^2 &> \frac{GMm}{R} \\ \Rightarrow v^2 &> \frac{2GM}{R} \\ \Rightarrow v &> \sqrt{\frac{2GM}{R}} \end{aligned}$$

which is approximately 10kms^{-1} for Earth.

Note: the mass cancels in v_{escape} because the *gravitational mass* (from the inverse square law) is equal to the *inertial mass* (that appears in Newton's second).

Proposition 2.4 (Principle of equivalence)

Objects have the same gravitational and inertial masses.

Remark. This is not known and there is no reason why we would expect this to be true – whether there may be tiny deviations between the masses is under investigation.

It's useful to write energy as $E = T + V$, where T is the *kinetic energy* $\frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}$.

Definition 2.5. The *work done* by a force as a particle moves along a trajectory C is given by

$$W = \int_C \mathbf{F} \cdot d\mathbf{x}$$

Proposition 2.6

For a conservative force, the work done along a trajectory only depends on the endpoints of the trajectory, not on the path itself.

Proof 1. We have

$$W = \int_C \mathbf{F} \cdot d\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F} \cdot \frac{d\mathbf{x}}{dt} dt$$

